Sistem Informasi Dalam Organisasi

Departemen Informatika
Fakultas Sains dan Matematika
Universitas Diponegoro

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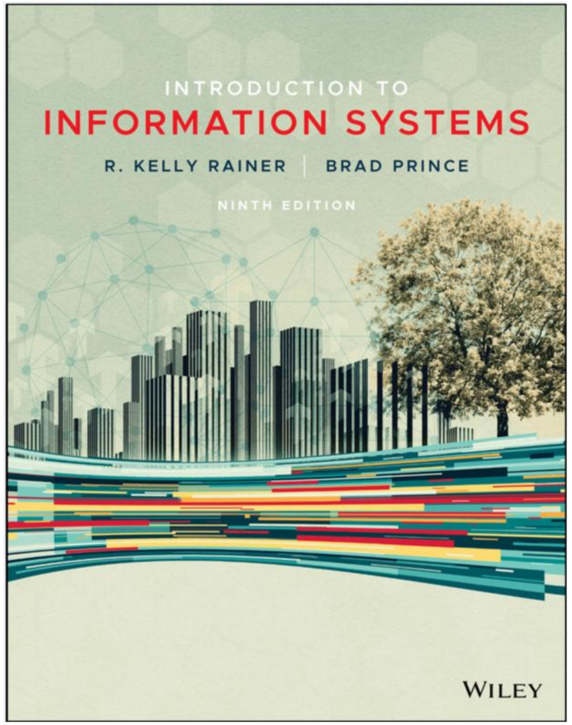


Chapter 10
Information Systems Within the organization

1. Transaction Processing Systems
2. Functional Area Information Systems
3. Enterprise Resource Planning (ERP) Systems
4. ERP Support for Business Process

ACCT **For the Accounting Major** highlights content relevant to the functional area of accounting.

FIN **For the Finance Major** highlights content relevant to the functional area of finance.

MKT **For the Marketing Major** highlights content relevant to the functional area of marketing.

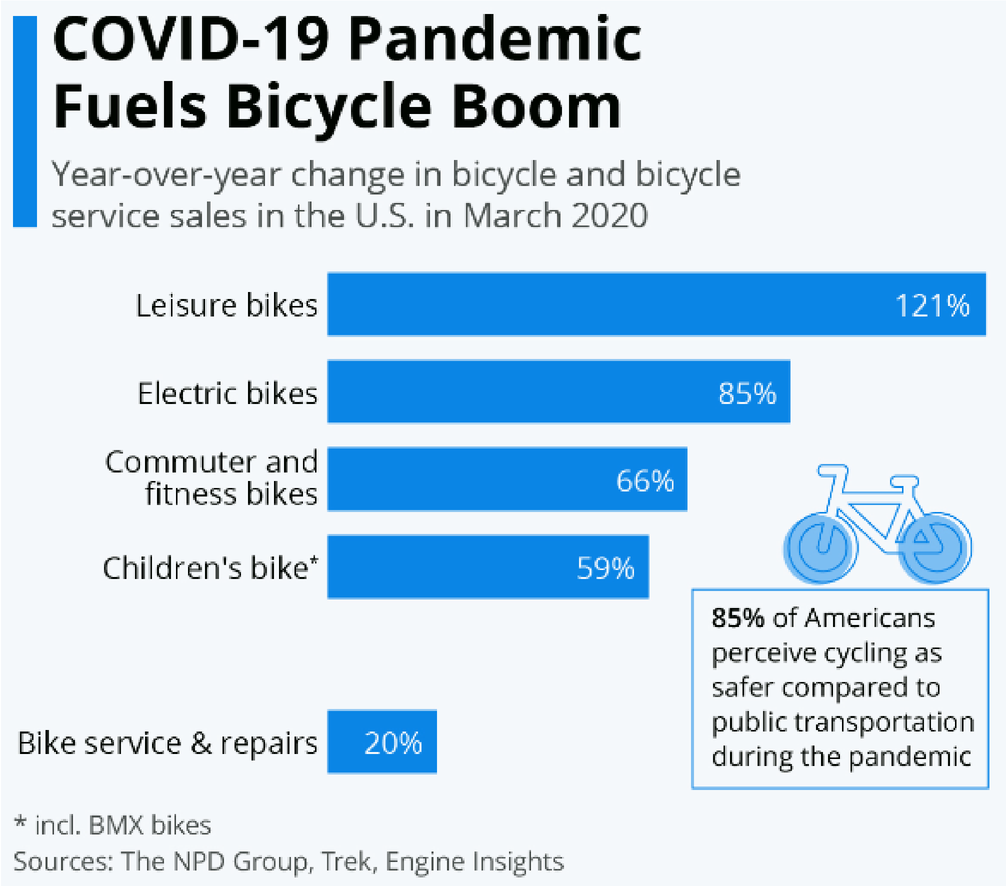
POM **For the Production/Operations Management Major** highlights content relevant to the functional area of production/operations management.

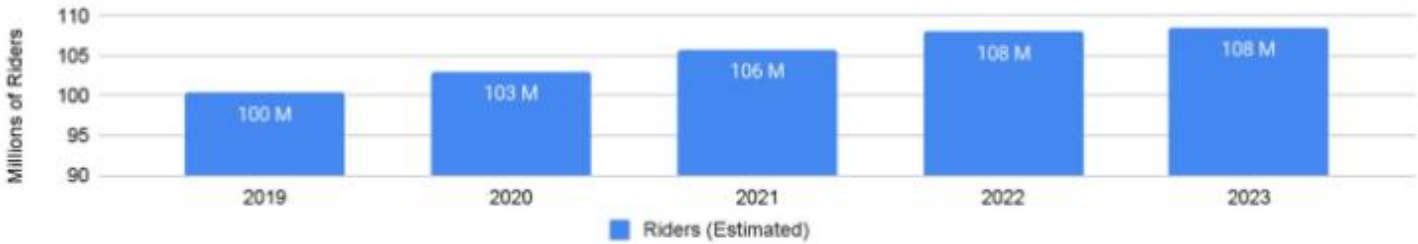
HRM **For the Human Resources Major** highlights content relevant to the functional area of human resources.

MIS **For the MIS Major** highlights content relevant to the functional area of MIS.

Opening Case

- The bike sale case discusses how the COVID-19 pandemic led to a surge in bicycle sales as people sought alternative transportation, resulting in inventory challenges for manufacturers like Trek, who strategically increased orders and maintained stock to meet the rising demand





https://communitycycles.org/rays-soapbox-covid-and-the-bike-biz-are-we-done-yet/

Transaction Processing Systems

- Transaction processing system (TPS) supports the monitoring, collection, storage, and processing of data from the organization's basic business transactions, each of which generates data.

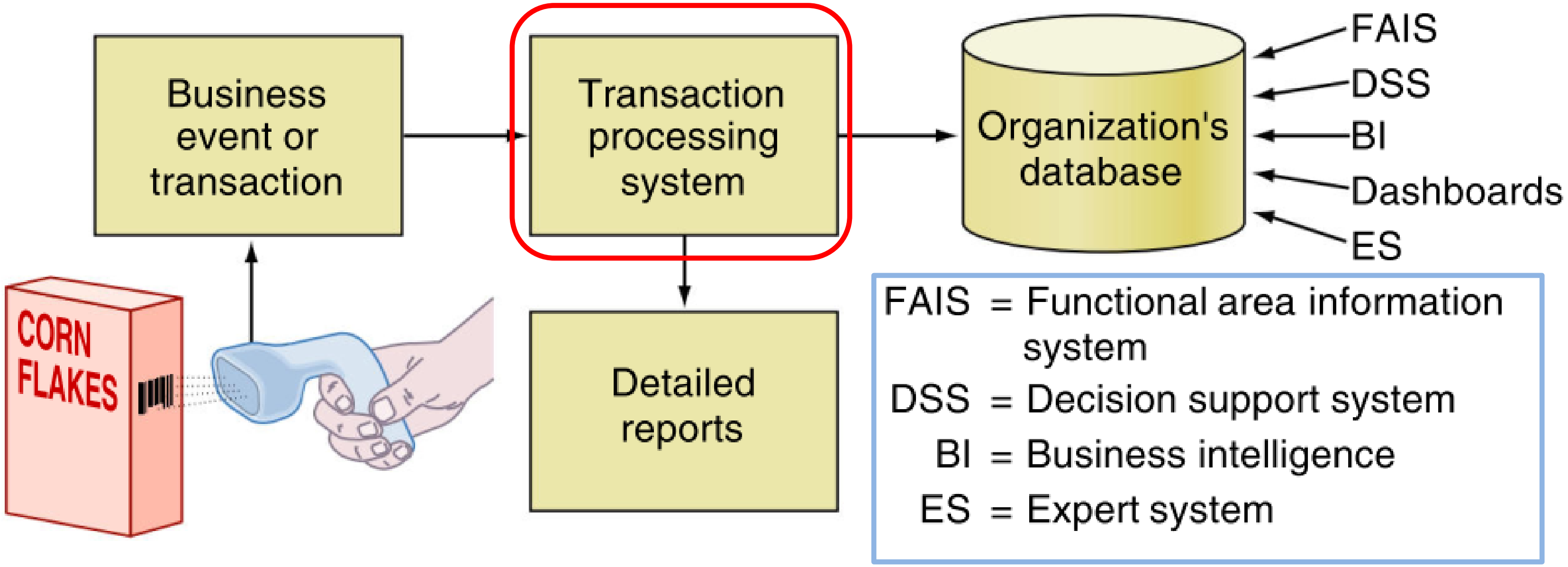
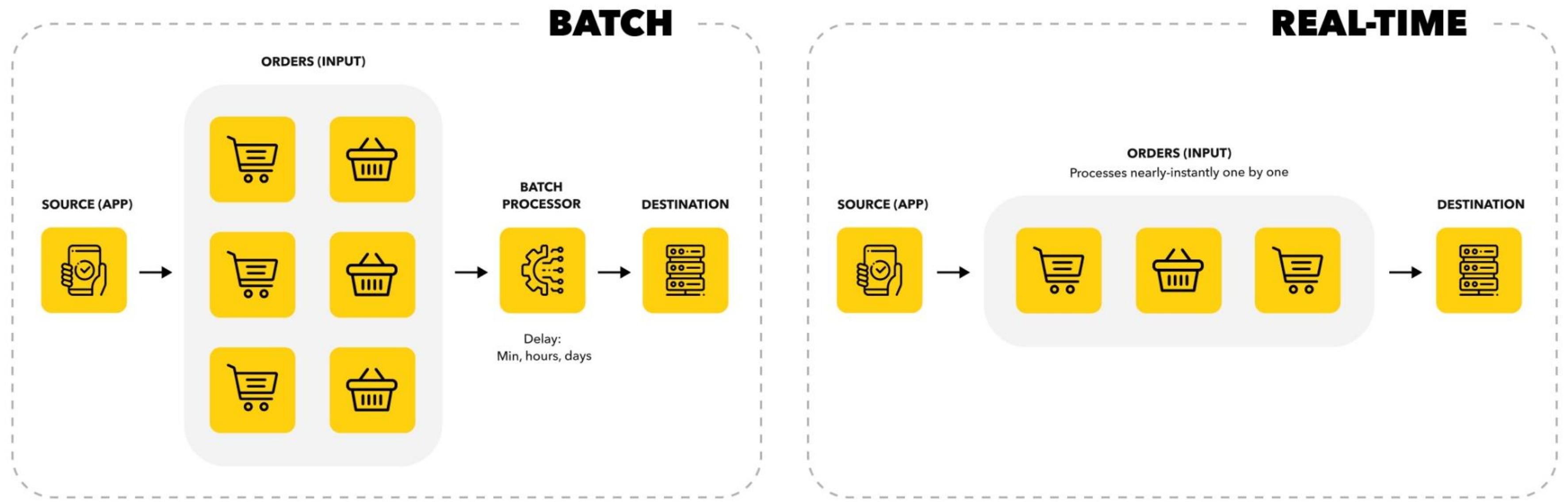


FIGURE 10.1 **How transaction processing systems manage data.**

Transaction Processing Systems

- Batch processing, the firm collects data from transactions as they occur, placing them in groups, or batches. The system then prepares and processes the batches periodically (say, every night)
- Online transaction processing (OLTP), business transactions are processed online as soon as they occur



Review

1. Define TPS.
2. List the key functions of a TPS.

Functional Area Information Systems

- Each department or functional area within an organization has its own collection of application programs, or information systems
- Functional area information systems (FAIS) supports a particular functional area in the organization by increasing each area's internal efficiency and effectiveness

- Examples of FAISs are:
  - √ accounting IS,
  - √ finance IS,
  - √ production/operations management (POM) IS,
  - √ marketing IS,
  - √ human resources IS

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  - √ marketing IS,
  - √ human resources IS

Examples of IS supporting the FA

	ACCOUNTING	FINANCE	HUMAN RESOURCES	PRODUCTION/ OPERATIONS	MARKETING
STRATEGIC	Profitability Planning	Financial Planning	Employment Planning, Outsourcing	Product Life Cycle Management	Sales Forecasting, Advertising Planning
TACTICAL	Auditing, Budgeting	Investment Management	Benefits Administration, Performance Evaluation	Quality Control, Inventory Management	Customer Relations, Sales Force Automation
OPERATIONAL	Payroll, Accounts Payable, Accounts Receivable	Manage Cash, Manage Financial Transactions	Maintain Employee Records	Order Fulfillment, Order Processing	Set Pricing, Profile Customers

FIGURE 10.3 **Examples of information systems supporting the functional areas.**

TABLE 10.1	Activities Supported by Functional Area Information Systems

ACCT FIN Accounting and Finance

Financial planning and cost of money

Budgeting—allocates financial resources among participants and activities

Capital budgeting—financing of asset acquisitions

Managing financial transactions

Handling multiple currencies

Virtual close—the ability to close the books at any time on short notice

Investment management—managing organizational investments in stocks, bonds, real estate, and other investment vehicles

Budgetary control—monitoring expenditures and comparing them against the budget

Auditing—ensuring the accuracy of the organization's financial transactions and assessing the condition of the organization's financial health

Payroll

MKT Marketing and Sales

Customer relations—knowing who customers are and treating them appropriately

Customer profiles and preferences

Salesforce automation—using software to automate the business tasks of sales, thereby improving the productivity of salespeople

POM Production/Operations and Logistics

Inventory management—when to order new inventory, how much inventory to order, and how much inventory to keep in stock

Quality control—controlling for defects in incoming materials and goods produced

Materials requirements planning (MRP)—planning process that integrates production, purchasing, and inventory management of interdependent items

Manufacturing resource planning (MRP II)—planning process that integrates an enterprise's production, inventory management, purchasing, financing, and labor activities

Just-in-time systems (JIT)—a principle of production and inventory control in which materials and parts arrive precisely when and where needed for production

Computer-integrated manufacturing—a manufacturing approach that integrates several computerized systems, such as computer-assisted design (CAD), computer-assisted manufacturing (CAM), MRP, and JIT

Product life cycle management—business strategy that enables manufacturers to collaborate on product design and development efforts, using the Web

HRM Human Resource Management

Recruitment—finding employees, testing them, and deciding which ones to hire

Performance evaluation—periodic evaluation by superiors

Training

Employee records

Benefits administration—retirement, disability, unemployment, and so on

Product Life Cycle

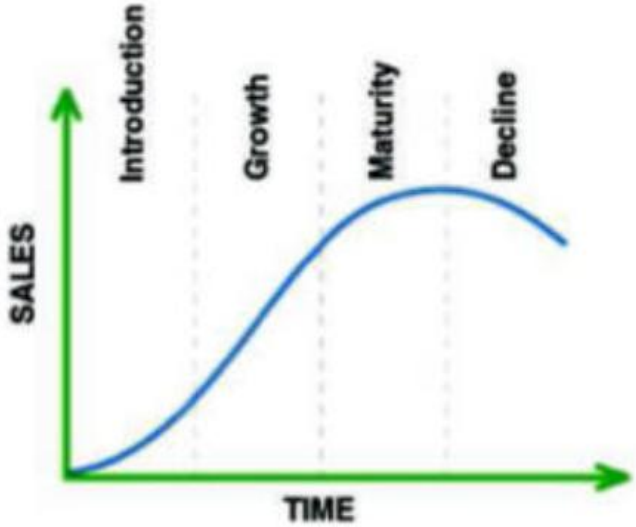


FIGURE 10.2 Product life cycle.

Reports

- All information systems produce reports: transaction processing systems, functional area information systems, ERP systems, customer relationship management systems, business intelligence systems, and so on.
- Three categories of reports are:
  - Routine
  - Ad hoc (on-demand)
  - Exception

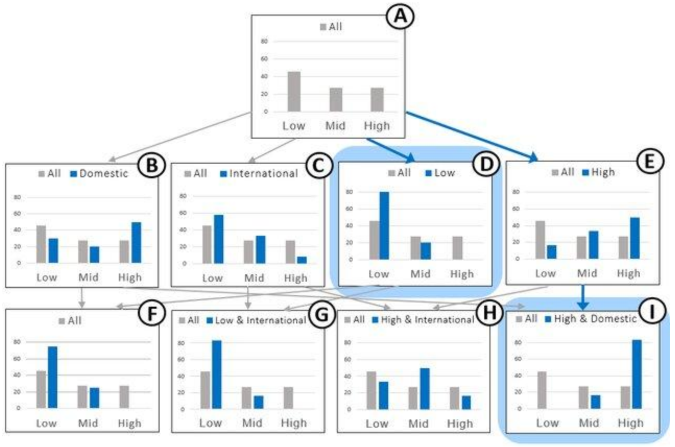
Reports app: Tableau vs Power BI

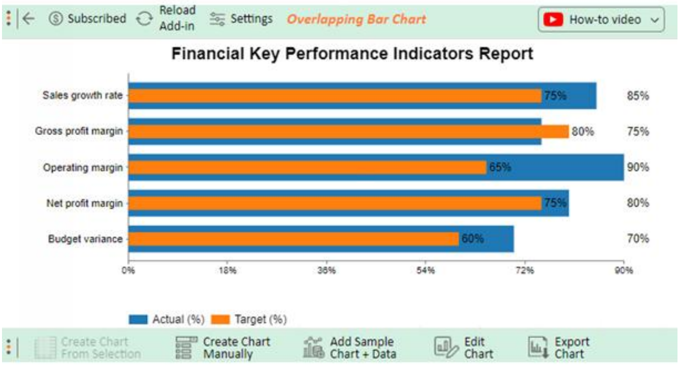
Routine Reports

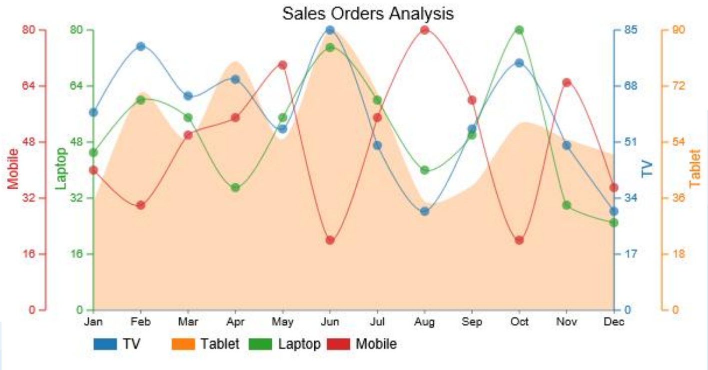
- Routine reports are produced at scheduled intervals.
- They range from hourly quality control reports to daily reports on absenteeism rates.

Ad hoc (on-demand) report

- Such out-of-the routine reports are called ad hoc (on-demand) reports.
- Ad hoc reports can also include requests for the following types of information:
  - Drill-down reports display a greater level of detail. For example, a manager might examine sales by region and decide to "drill down" by focusing specifically on sales by store and then by salesperson.
  - Key indicator reports summarize the performance of critical activities. For example, a chief financial officer might want to monitor cash flow and cash on hand.
  - Comparative reports compare, for example, the performances of different business units or of a single unit during different times.







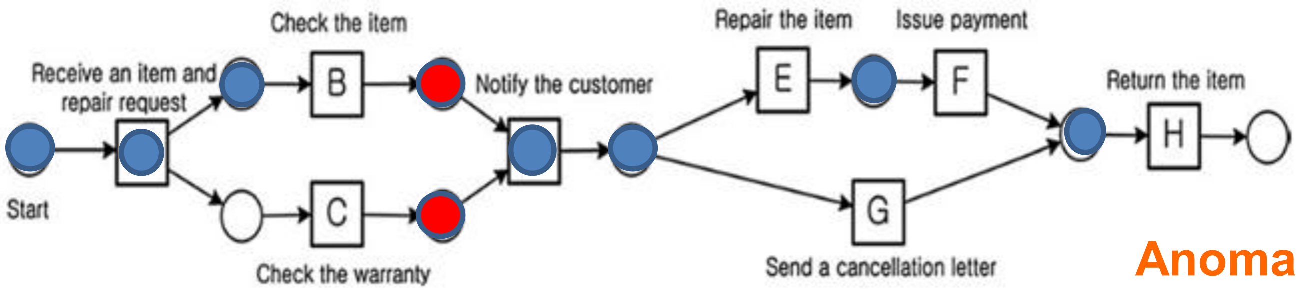
Exception Reports

- Exception reports include only information that falls outside certain threshold standards.
- To implement management by exception, management first establishes performance standards.
- The company then creates systems to monitor performance (through the incoming data about business transactions such as expenditures), to compare actual performance to the standards, and to identify exceptions to the standards.
- The system alerts managers to the exceptions through exception reports

Exception Reports Mechanism
Example: Conformance Checking

- Deteksi Anomali menggunakan perspektif control flow (proses)
  - Case id 1 dan case id 4 mengandung anomali
  - Case id 2 dan case id 3 normal

Case ID:1
- Receive an item and repair request (A)
- Notify the customer (D)
- Check the item (B)
- Repair the item (E)
- Issue payment (F)
- Return the item (H)



Anomali

Tampilan pada layar monitoring

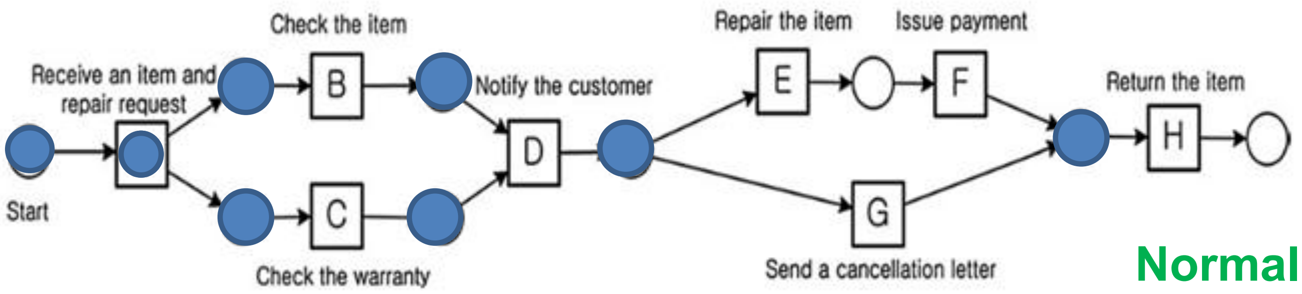
waktu

```
case id:  1 Start
case id:  2 Start
>>ALERT! [ 1 ][Anomaly score: 1.0, accu scores: 1.0 ] [ Notify the customer ] [Type: missing token] Sue
>>>WARNING! AN INSPECTION NEEDED ON CASE ID: 1
case id:  3 Start
case id:  4 Start
>>ALERT! [ 4 ][Anomaly score: 1.0, accu scores: 1.0 ] [ Send a cancellation letter ] [Type: missing token] Clare
>>>WARNING! AN INSPECTION NEEDED ON CASE ID: 4
```

Exception Reports
Example: Conformance Checking

- Deteksi Anomali menggunakan perspektif control flow (proses)
  - Case id 1 dan case id 4 mengandung anomali
  - Case id 2 dan case id 3 normal

Case ID: 2 (perspektif tunggal)
- Receive an item and repair request (A)
- Check the waranty (C)
- Check the item (B)
- Notify the customer (D)
- Send a cancellation letter (G)
- Return the item (H)



Normal

Tampilan pada layar monitoring

waktu

```
case id:  1 Start
case id:  2 Start
>>ALERT! [ 1 ][Anomaly score: 1.0, accu scores: 1.0 ] [ Notify the customer ] [Type: missing token] Sue
>>>WARNING! AN INSPECTION NEEDED ON CASE ID: 1
case id:  3 Start
case id:  4 Start
>>ALERT! [ 4 ][Anomaly score: 1.0, accu scores: 1.0 ] [ Send a cancellation letter ] [Type: missing token] Clare
>>>WARNING! AN INSPECTION NEEDED ON CASE ID: 4
```

Review

1. Define a functional area information system and list its major characteristics.
2. How do information systems benefit the finance and accounting functional area?
3. Explain how POM personnel use information systems to perform their jobs more effectively and efficiently.
4. What are the most important HRIS applications?
5. Compare and contrast the three basic types of reports.

Break

Sistem Informasi Dalam Organisasi

Enterprise Resource Planning (ERP) Systems

- Enterprise resource planning (ERP) systems are designed to correct a lack of communication among the FAISs.

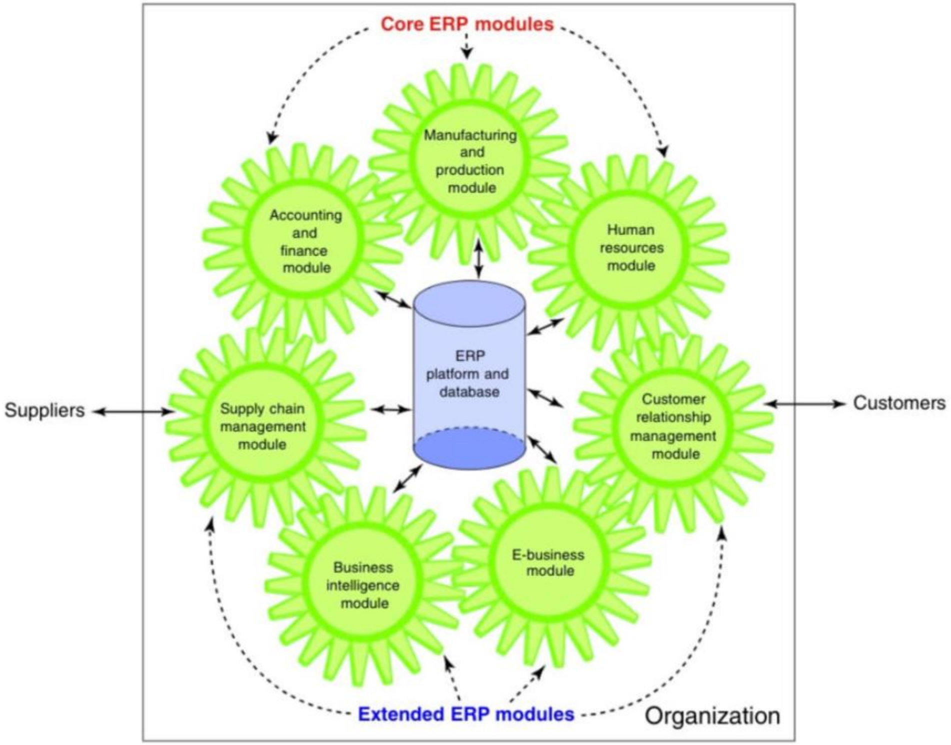
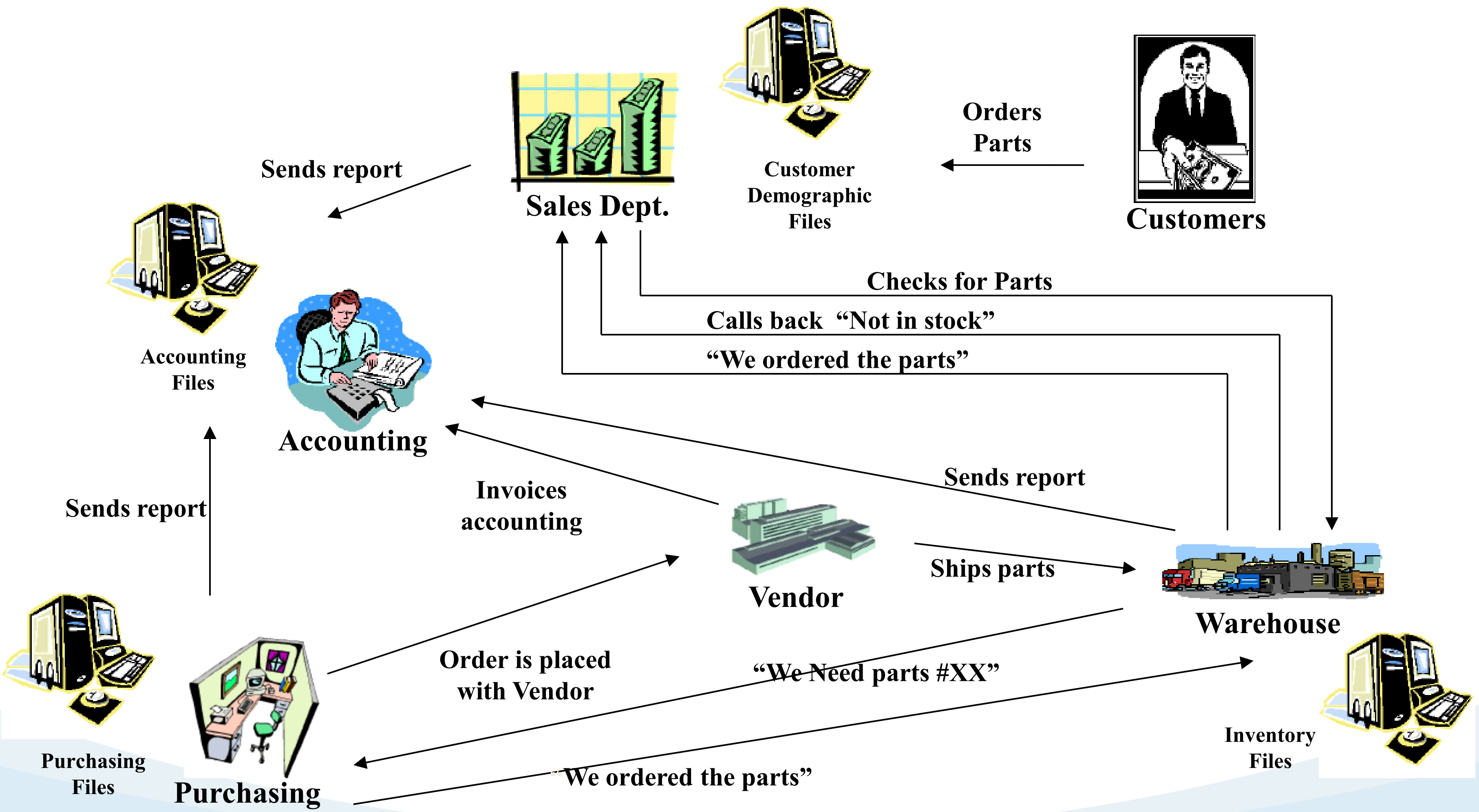


FIGURE 10.4 ERP II system.

Aspect	TPS	ERP
Focus	Daily transaction processing	Integration across business functions
Purpose	Manage operational transactions	Improve efficiency and decision-making
Scale	Typically used at the operational level	Used by the entire organization
Users	Operational staff	Management and various departments
Example Activities	Sales, order entry, payments	Resource management, strategic planning

Before ERP



Sales Dept.
Customer
Demographic
Files
Orders
Parts
Customers
Sends report
Accounting
Files
Accounting
Checks for Parts
Calls back "Not in stock"
"We ordered the parts"
Sends report
Invoices
accounting
Sends report
Vendor
Ships parts
Warehouse
Order is placed
with Vendor
"We Need parts #XX"
Purchasing
Files
Purchasing
"We ordered the parts"
Inventory
Files

After ERP

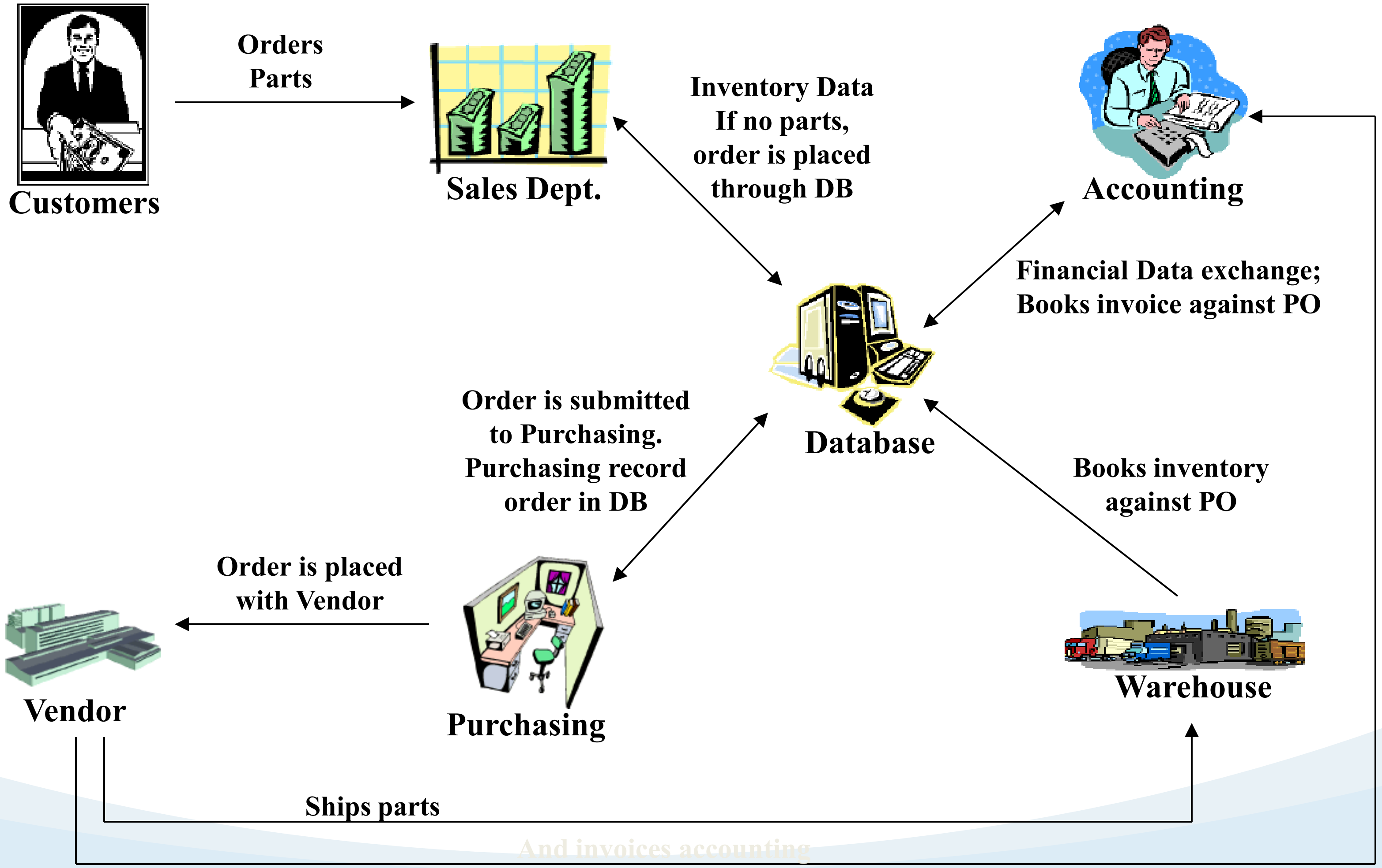

Customers
Orders
Parts
Sales Dept.
Inventory Data
If no parts,
order is placed
through DB
Accounting
Financial Data exchange;
Books invoice against PO
Database
Order is submitted
to Purchasing.
Purchasing record
order in DB
Books inventory
against PO
Order is placed
with Vendor
Vendor
Purchasing
Warehouse
Ships parts
And invoices accounting

TABLE 10.2 **ERP Modules**

Core ERP Modules
ACCT FIN **Financial Management.** These modules support accounting, financial reporting, performance management, and corporate governance. They manage accounting data and financial processes such as general ledger, accounts payable, accounts receivable, fixed assets, cash management and forecasting, product-cost accounting, cost-center accounting, asset accounting, tax accounting, credit management, budgeting, and asset management.
POM **Operations Management.** These modules manage the various aspects of production planning and execution such as demand forecasting, procurement, inventory management, materials purchasing, shipping, production planning, production scheduling, materials requirements planning, quality control, distribution, transportation, and plant and equipment maintenance.
HRM **Human Resource Management.** These modules support personnel administration (including workforce planning, employee recruitment, assignment tracking, personnel planning and development, and performance management and reviews), time accounting, payroll, compensation, benefits accounting, and regulatory requirements.

Extended ERP Modules

MKT **Customer Relationship Management.** (Discussed in detail in Chapter 11.) These modules support all aspects of a customer's relationship with the organization. They help the organization to increase customer loyalty and retention, and thus improve its profitability. They also provide an integrated view of customer data and interactions, helping organizations to be more responsive to customer needs.

POM **Supply Chain Management.** (Discussed in detail in Chapter 11.) These modules manage the information flows between and among stages in a supply chain to maximize supply chain efficiency and effectiveness. They help organizations plan, schedule, control, and optimize the supply chain from the acquisition of raw materials to the receipt of finished goods by customers.

MIS **Business Analytics.** (Discussed in detail in Chapter 12.) These modules collect information used throughout the organization, organize it, and apply analytical tools to assist managers with decision making.

MIS **E-Business.** (Discussed in detail in Chapter 7.) Customers and suppliers demand access to ERP information, including order status, inventory levels, and invoice reconciliation. Furthermore, they want this information in a simplified format that can be accessed on the Web. As a result, these modules provide two channels of access into ERP system information—one channel for customers (B2C) and one for suppliers and partners (B2B).

Benefits of ERP Systems

1. Organizational Flexibility And agility: ERP systems break down many former departmental and functional silos of business processes, information systems, and information resources.
   - √ In this way, they make organizations more flexible, agile, and adaptive.
   - √ The organizations can therefore respond quickly to changing business conditions and capitalize on new business opportunities.

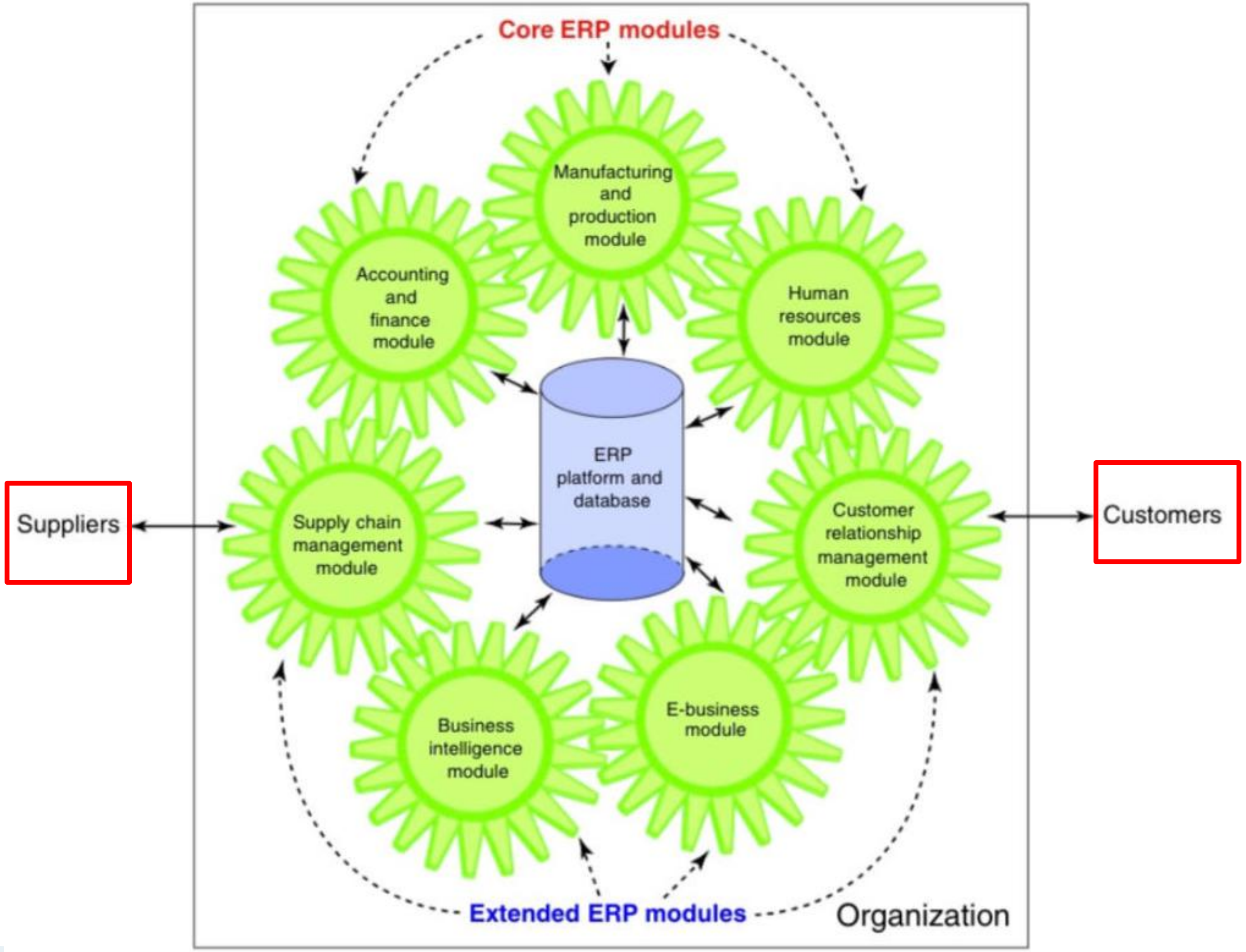


FIGURE 10.4 ERP II system.

Benefits of ERP Systems

2. Decision support: ERP systems provide essential information on business performance across functional areas. This information significantly improves managers' ability to make better, more timely decisions.
3. Quality and efficiency: ERP systems integrate and improve an organization's **business processes**, generating significant improvements in the quality of production, distribution, and customer service.

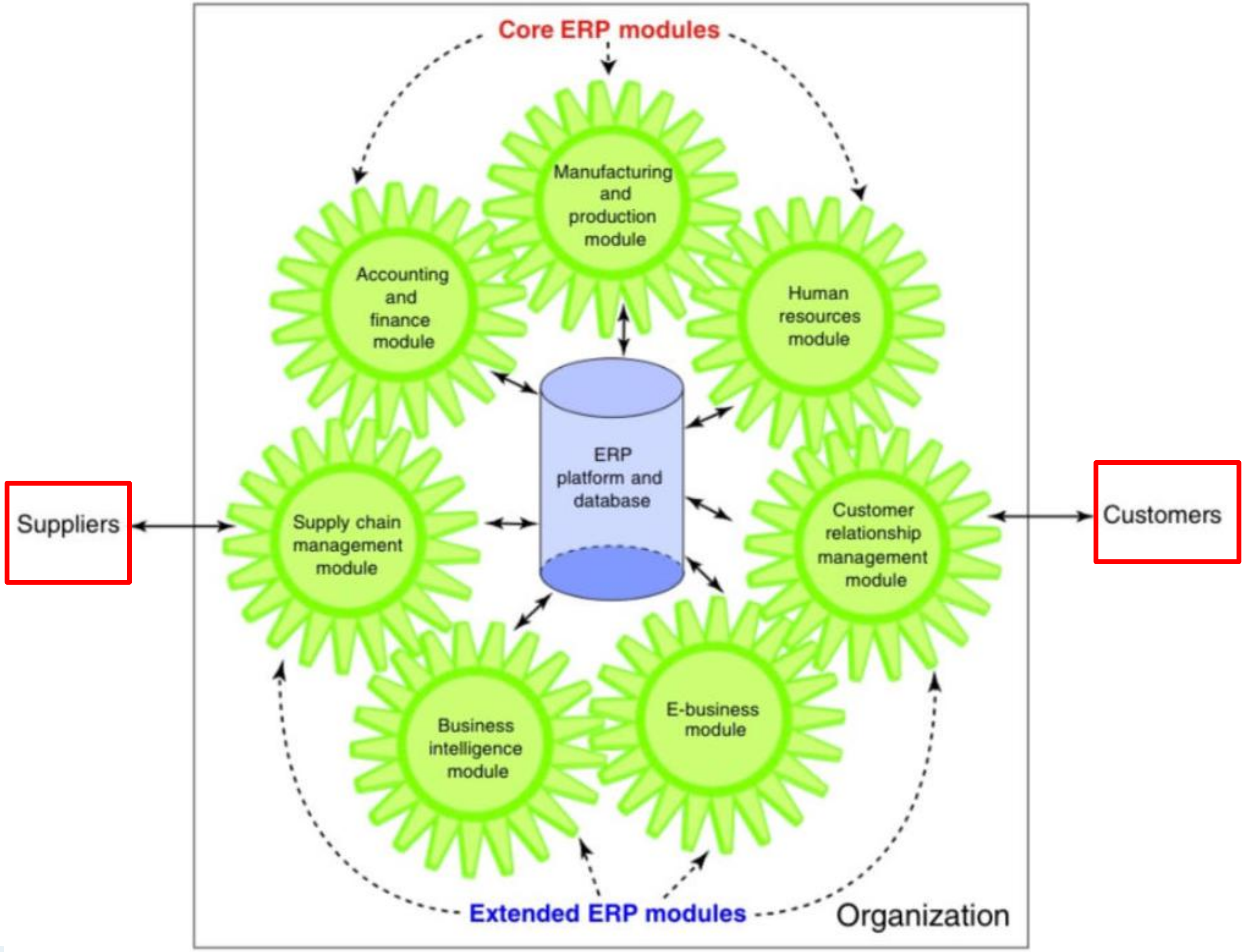


FIGURE 10.4 ERP II system.

Limitations of ERP Systems

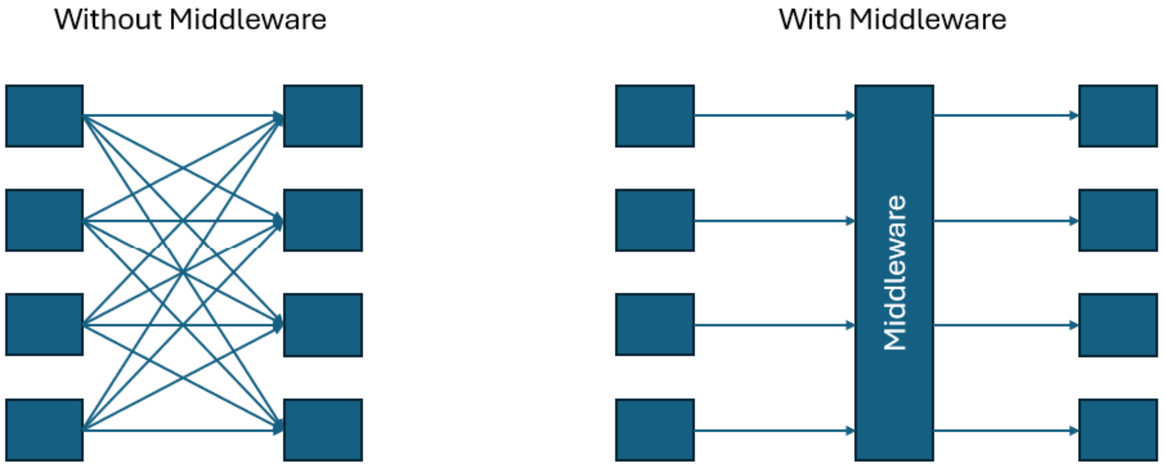
- ERP Systems can be extremely complex, expensive, and time consuming to implement

Implementing ERP Systems

- On-Premise ERP Implementations
  - The vanilla approach
  - The custom approach
  - The best-of-breed approach
- Software-as-a-Service ERP Implementation, advantages:
  - √ Any location access
  - √ Low investments
  - √ Scalable
- SaaS major disadvantages
  - Security reason
  - Lack of control over the IT resources → Business Continuity problems

Enterprise Application Integration

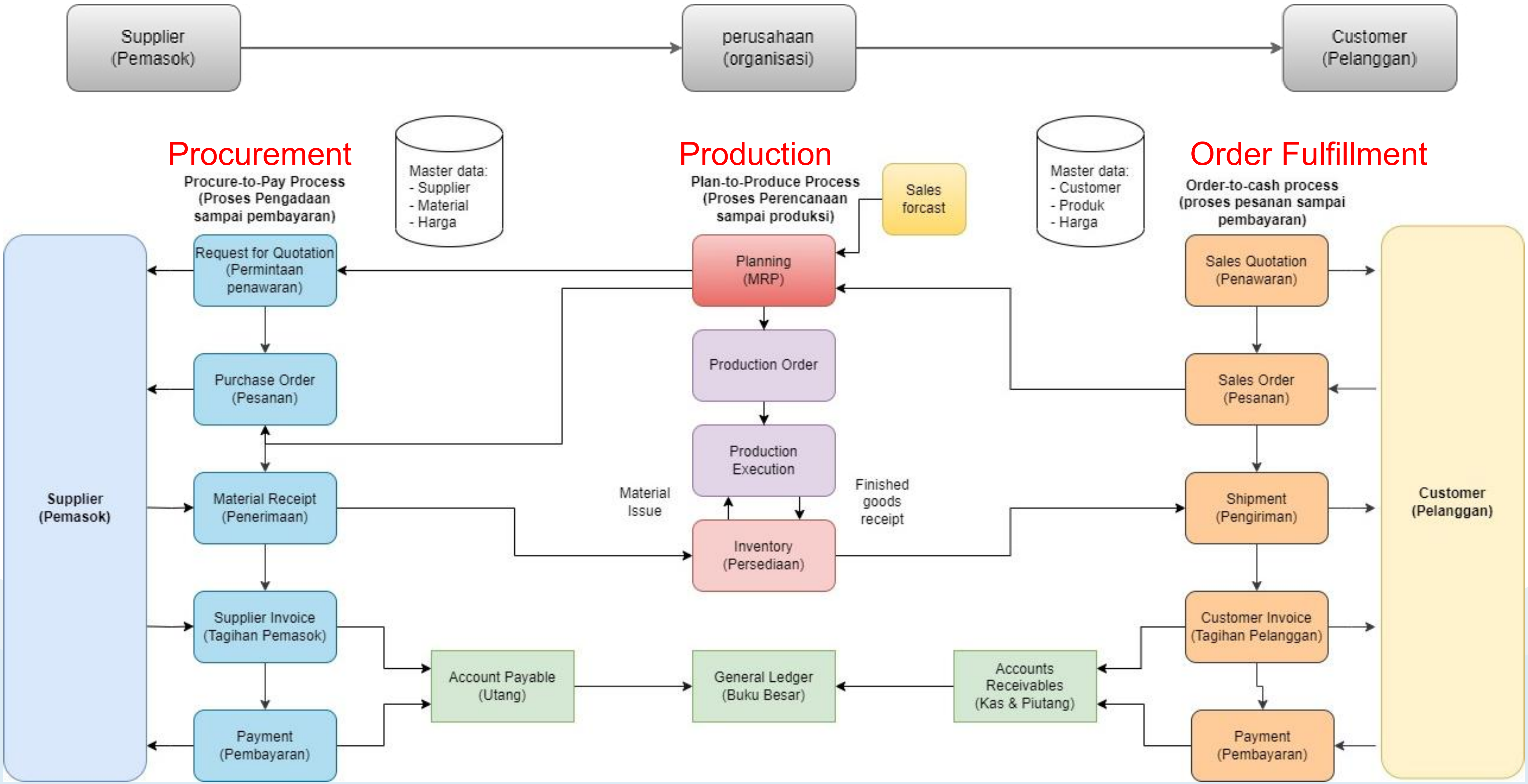
- Enterprise application integration (EAI) system integrates existing systems by providing software, called middleware, that connects multiple applications



Review

1. Define ERP and describe its functions.
2. What are ERP II systems?
3. Differentiate between core ERP modules and extended ERP modules.
4. List some drawbacks of ERP software.
5. Highlight the differences between ERP configuration, customization, and best-of-breed implementation strategies.

Main Business Process in ERP



ERP Support for Business Processes

- ERP systems effectively support a number of standard business processes. ERP Systems manage end-to-end, cross – departmental processes.
  - Procurement Process
  - Order Fulfillment Process
  - Production Process

Procurement Process

- The procurement process, which originates in the warehouse department (need to buy) and ends in the accounting department (send payment)

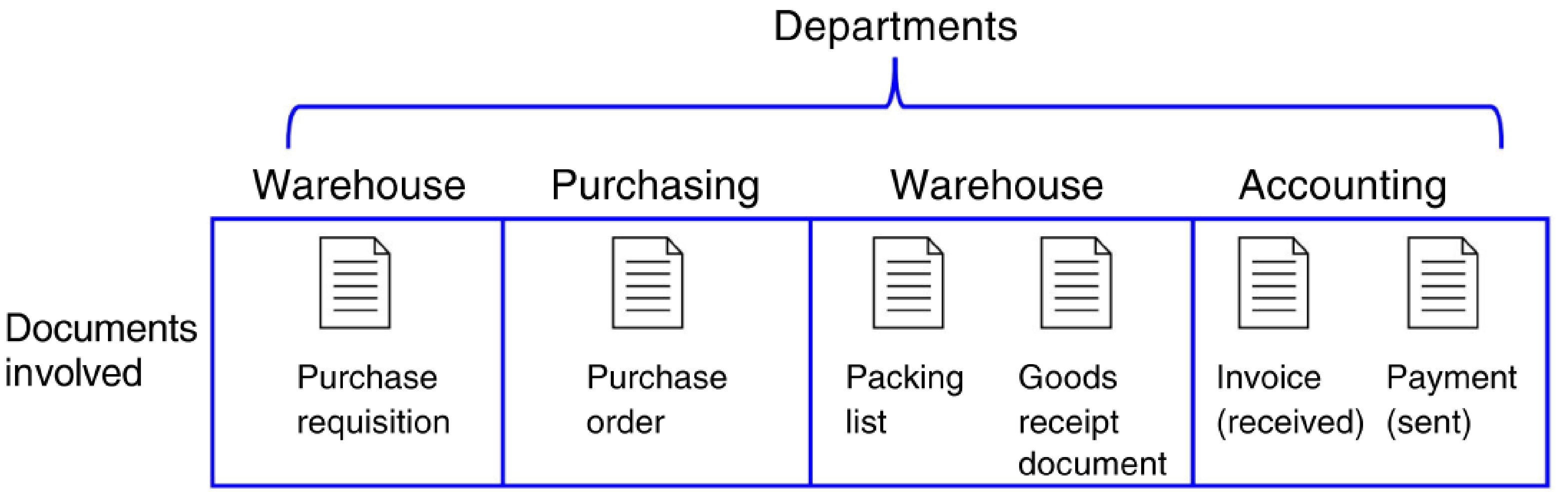
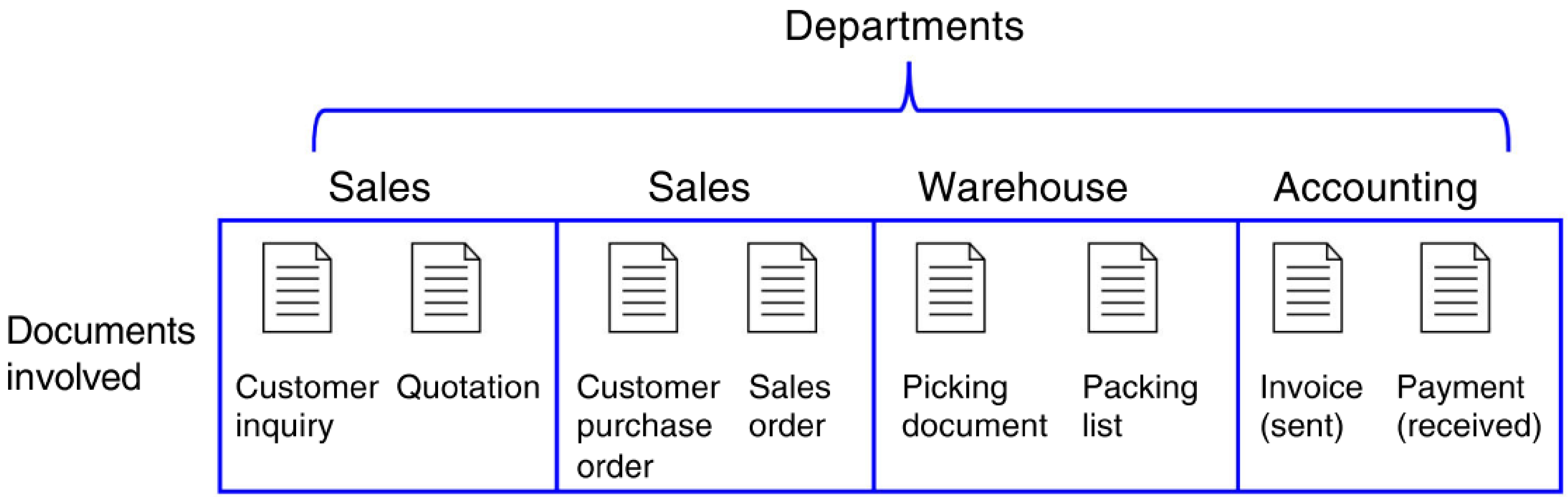


FIGURE 10.5 **Departments and documents flow in the procurement process.**

Fulfillment Process

- The fulfillment process, which originates in the sales department (customer request to buy) and ends in the accounting department (receive payment)

FIGURE 10.6 Departments and documents flow in the fulfillment process.



Production Process

- The production process, which originates and ends in the warehouse department (need to produce and reception of finished goods) but involves the production department as well

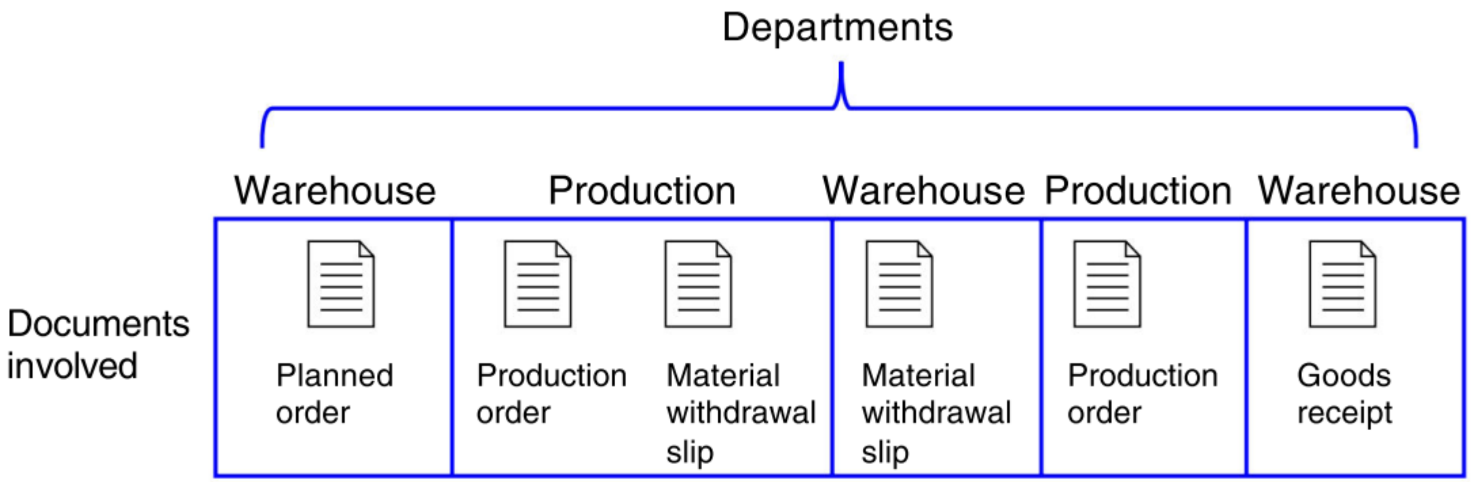


FIGURE 10.7 **Departments and documents flow in the production process.**

Integrated Processes with ERP Systems

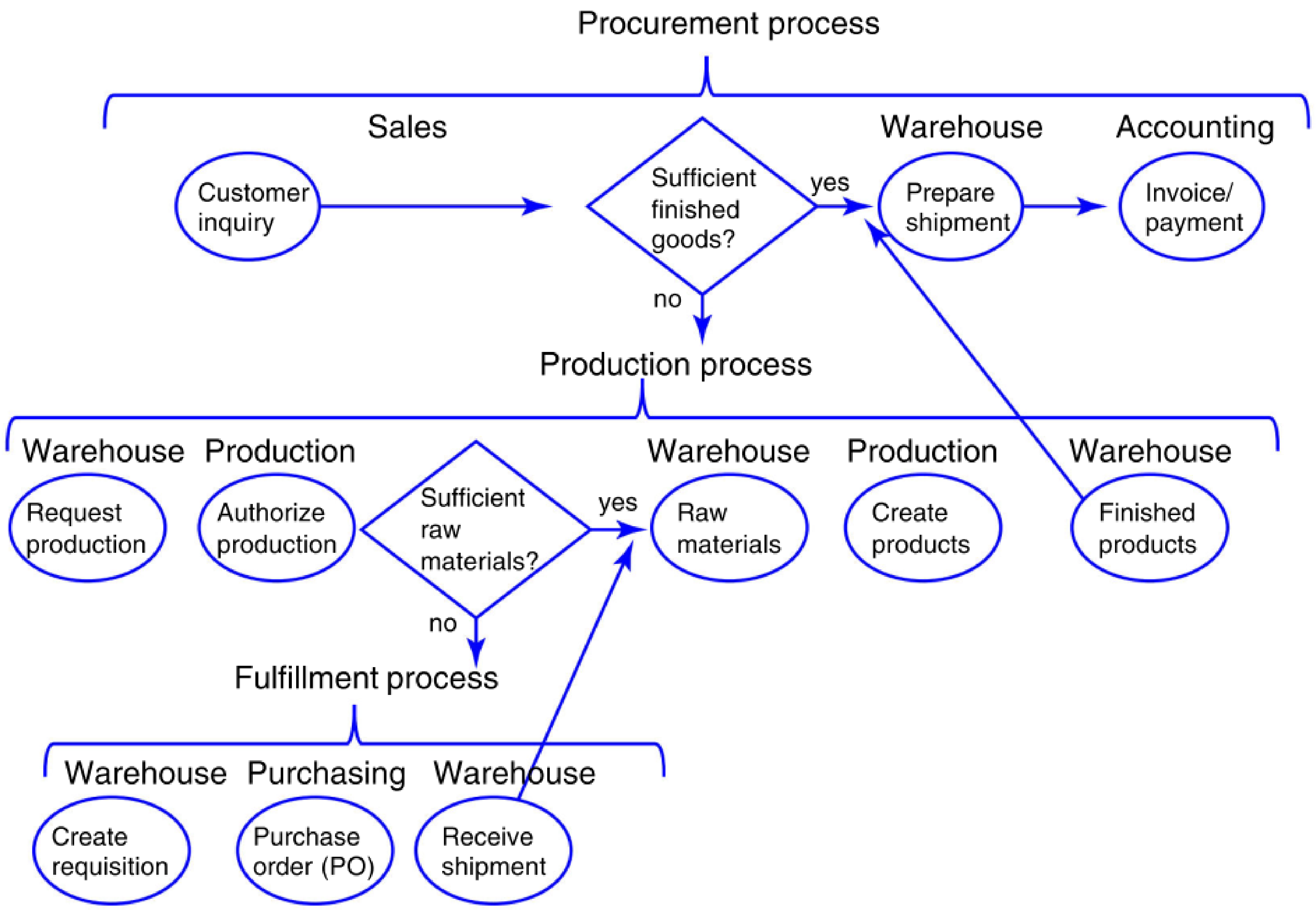


FIGURE 10.8 **Integrated processes with ERP systems.**

Inter-Organizational Processes: ERP with SCM and CRM

- **Procurement** and **fulfillment** processes are considered intra-organizational processes because they originate and conclude within the company
- Inter-organizational Processes manage processes that originate in one company and conclude in another company:
  - ❑ Supply Chain Management (SCM)
  - ❑ Costumer Relationship Management (CRM)

ERP SCM

- ERP SCM systems have the capability to place automatic requests to buy fresh perishable products from suppliers in real time.
  - That is, as each perishable product is purchased, the system captures data on that purchase, adjusts store inventory levels, and transmits these data to the grocery chain's warehouse as well as the products' vendors.
  - The system executes this process by connecting the point-of-sale barcode scanning system with the warehouse and accounting departments, as well as with the vendors' systems.
  - SCM systems also use historical data to predict when fresh products need to be ordered before the store's supply becomes too low.

https://www.forbes.com/sites/moorinsights/2024/02/06/erp-and-scm---what-to-expect-in-2024-part-2/

ERP CRM

- ERP CRM systems also benefit businesses by generating forecasting analyses of product consumption based on critical variables such as geographical area, season, day of the week, and type of customer.
  - ➢ These analyses help grocery stores coordinate their supply chains to meet customer needs for perishable products.
  - ➢ Going further, CRM systems identify particular customer needs and then use this information to suggest specific product campaigns.
  - ➢ These campaigns can transform a potential demand into sales opportunities and convert sales opportunities into sales quotations and sales orders.
  - ➢ This process is called the demand-to-order process.

Review

1. What are the three main intraorganizational processes that are typically supported by ERP systems?
2. Why is it important that all steps in each process generate a document that is stored in the ERP system?
3. What is the difference between intraorganizational and interorganizational processes?
4. What are the two main ES systems that support interorganizational processes?

Discussion

1. Why is it logical to organize IT applications by functional areas?
2. Describe the role of a TPS in a service organization.
3. Describe the relationship between TPS and FAIS.
4. Discuss how IT facilitates the budgeting process.
5. How can the Internet support investment decisions?
6. Describe the benefits of integrated accounting software packages.
7. Discuss the role that IT plays in support of auditing.
8. Investigate the role of the Web in human resources management.
9. What is the relationship between information silos and enterprise resource planning?

Summary

10.1 Explain the purpose of transaction processing systems.

TPSs monitor, store, collect, and process data generated from all business transactions. These data provide the inputs into the organization's database.

10.2 Explain the types of support that information systems can provide for each functional area of the organization.

The major business functional areas are production/operations management, marketing, accounting/finance, and human resources management. Table 10.1 provides an overview of the many activities in each functional area supported by FAIS.

10.3 Identify advantages and drawbacks to businesses of implementing an ERP system.

Enterprise resource planning (ERP) systems integrate the planning, management, and use of all of the organization's resources. The major objective of ERP systems is to tightly integrate the functional areas of the organization. This integration enables information to flow seamlessly across the various functional areas.

The following are the major benefits of ERP systems.

- Because ERP systems integrate organizational resources, they make organizations more flexible, agile, and adaptive. The organizations can therefore react quickly to changing business conditions and capitalize on new business opportunities.
- ERP systems provide essential information on business performance across functional areas. This information significantly improves managers' ability to make better, more timely decisions.
- ERP systems integrate organizational resources, resulting in significant improvements in the quality of customer service, production, and distribution.

The following are the major drawbacks of ERP systems.

- The business processes in ERP software are often predefined by the best practices that the ERP vendor has developed. As a result, companies may need to change existing business processes to fit the predefined business processes of the software. For companies with well-established procedures, this requirement can be a huge problem.
- ERP systems can be extremely complex, expensive, and time consuming to implement. In fact, the costs and risks of failure in implementing a new ERP system are substantial.

10.4 Describe the three main business processes supported by ERP systems.

The *procurement process*, which originates in the warehouse department (need to buy) and ends in the accounting department (send payment).

The *fulfillment process* that originates in the sales department (customer request to buy) and ends in the accounting department (receive payment).

The *production process* that originates and ends in the warehouse department (need to produce and reception of finished goods), but involves the production department as well.

We leave the details of the steps in each of these processes up to you.

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